

Granulation-Based Temporal Convolutional Network for Student Engagement Detection in E-Learning Environments

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Abstract

Something important disappears when teaching moves online, and instructors rarely notice it until a session goes wrong: the low-bandwidth but continuous read of how a room is actually doing. A puzzled look, a student pushing back in their chair, eyes drifting to a second screen — none of this registers in click logs or quiz completion rates, and by the time those records update the moment to intervene has already closed. GranulationCNN is a webcam-based engagement estimator whose design starts from one observation: averaging a ten-second clip into a single embedding vector throws away information that probably matters. Within a single ten-second window a student might look at the screen, glance somewhere else, and refocus — a mean over those frames smooths the variation into invisibility. Our Temporal Granulation Module (TGM) takes a different route: it randomly partitions the frame embeddings into several groups, computes a mean prototype for each group, and aggregates the resulting per-granule logits into a single prediction. Across training iterations the partition keeps changing, which functions as implicit regularisation; a compactness term prevents any given granule from becoming a meaningless scatter of embeddings.

Evaluation uses DAiSEE, scoped to the C2 (moderate) and C3 (high) engagement levels — C0 and C1 together cover under 5% of the test split, too few for any standard cross-entropy objective to learn reliable representations. With an ImageNet-pretrained backbone (Variant C), GranulationCNN achieves 56.15% accuracy and weighted F1 of 0.4587 on the binary C2/C3 evaluation (Cohen’s $\kappa = 0.0468$), with cross-seed variance confirming the result is not an artefact: Acc 0.530 ± 0.004 , κ 0.038 ± 0.007 over three runs. A three-variant ablation — the main empirical contribution — shows that random initialisation yields $\kappa \approx 0$ regardless of seed, that pretraining is the single change that makes training converge reliably, and that focal loss trades aggregate accuracy for minority-class recall in a predictable way. The model is framed throughout as a decision-support signal for instructors, not an autonomous monitor making high-stakes calls about individual learners.

Keywords: student engagement detection; temporal granulation; affective computing; adaptive e-learning; decision support systems

1 Introduction

Remote and hybrid instruction has become the default for a large fraction of university teaching, not a pandemic-era exception. MOOC platforms routinely serve hundreds of thousands of learners at once. The scale is real. What gets lost at scale is harder to quantify: the instructor’s ability to read the room.

In a physical classroom a teacher reads the room continuously. A student looks puzzled at a particular slide; another keeps glancing at the door; a third leans forward when the material gets interesting. None of this is precise, but it is fast, and a competent teacher uses it to adjust pace,

restart an explanation, or pause for a question. The online equivalent is mostly delayed traces: attendance logs, click streams, quiz attempts, time on task. These are scalable but lagging. They report what already happened (Papamitsiou and Economides, 2014; Siemens and Baker, 2012; Alario-Hoyos et al., 2016).

So the question this paper is actually trying to answer is not “can a model classify engagement accurately?” It is whether a Learning Management System can give an instructor something closer to a live signal — not a post-hoc summary. Engagement is a layered construct that spans behavioural participation, cognitive effort, and emotional connection to the activity (Fredricks et al., 2004; Carini et al., 2006). In video these dimensions appear indirectly, through gaze and expression and posture. A single frame is a weak unit of evidence — a glance away proves nothing. Ten-second clips are sturdier; DAiSEE is built around them (Gupta et al., 2016).

Most lightweight clip-level models still collapse those ten seconds into one representation: an average, a recurrent state, or an attention-weighted summary. Pooling is efficient. It is also, quietly, a strong modelling assumption: that the whole clip has one dominant behavioural state. For an engagement clip this is often just wrong. At the other end, full video transformers and multi-stream networks model temporal structure richly, but their compute requirements are incompatible with processing many concurrent learner streams in a live LMS.

GranulationCNN is our attempt at a middle path. The spatial encoder stays small — ResNet-18 — but instead of pooling all frame embeddings into one vector, the Temporal Granulation Module (TGM) partitions them stochastically into several granules, builds a prototype per granule, and averages the per-granule logits. The random partition changes at every forward pass, exposing the model to many views of the same clip across training; a compactness loss prevents the granules from becoming arbitrary noise groups.

Three things are contributed. First, temporal granulation as a compact multi-prototype alternative to global pooling for engagement video. Second, stochastic frame-to-granule assignment as a regularisation device, paired with a compactness term that ties it to embedding geometry. Third, a three-variant controlled ablation that answers a question prior work rarely addresses: what does each design choice actually buy? The short answer — random initialisation collapses, pretraining stabilises, focal loss recovers minority classes — is cleaner than we expected going in. One upfront note: we restrict evaluation to DAiSEE’s C2 and C3 classes throughout. C0 and C1 together are fewer than 5% of the test split, which is not enough to learn from. Severe disengagement is a real problem; it is just not this paper’s problem.

2 Literature Review and Research Gap

2.1 Learning analytics and decision support in e-learning

Learning Analytics and Educational Data Mining start from the same place: the traces learners leave behind in a system should be turned into something teachers, students, and designers can act on (Papamitsiou and Economides, 2014; Siemens and Baker, 2012; Baker and Inventado, 2014). In practice this means the LMS is a feedback loop, not a content shelf. It records what learners do, filters that record, and ideally surfaces something actionable — a pacing adjustment, a content recommendation, a nudge to a struggling student, an alert to an instructor (Power, 2002; Alario-Hoyos et al., 2016). Engagement estimation belongs in that loop. A platform that can flag dropping attention during a session gives the instructor time to act. A quiz result the next morning does not.

The difficulty is that engagement does not appear directly in the data an LMS collects automatically. Clicks, time-on-task, forum activity, quiz scores — all of these are scalable and easy to gather, but they are coarse and they arrive late. Video is the closest proxy to what a teacher sees in person, but only if the resulting signal is cheap enough to run for many concurrent learners, accurate enough to be worth showing, and legible enough that an instructor reads it

as evidence rather than an oracle verdict.

2.2 Student engagement and affective computing

Affective computing has built a plausible but qualified case that gaze, expression, posture, and head motion carry information about how a learner is doing — qualified because the signals are noisy, context-sensitive, and far from deterministic (D’Mello and Graesser, 2012). Early work used handcrafted features and conventional classifiers. Whitehill et al. (2014) demonstrated above-chance engagement estimation from facial features with an SVM, which settled the basic feasibility question.

Deep learning improved the representations. Nezami et al. (2019) moved to CNN-derived facial features in an online learning context; Zhang et al. (2019) added gaze and head pose. Richer inputs help, but they also require richer sensing. We stay with plain RGB frames from a standard webcam, not because it is easier but because that is the hardware learners already have.

2.3 Temporal video representation

Video understanding offers no shortage of ways to model time. Two-stream networks handle appearance and motion separately (Simonyan and Zisserman, 2014). Temporal Segment Networks sample sparsely to keep inference tractable (Wang et al., 2016). Non-local networks capture long-range temporal interactions (Wang et al., 2018). Transformer-based variants extend spatial attention across the time dimension (Vaswani et al., 2017; Bertasius et al., 2021). All of these are capable architectures. None of them are easy to run for twenty or thirty concurrent learner streams inside a live LMS.

Average pooling persists because it is cheap, stable, and trivially differentiable. What it trades away is the assumption that a clip has one behavioural state throughout. For engagement, that trade is often a bad one: within ten seconds a learner can drift and return, and a mean over those frames conflates the two states. GranulationCNN accepts the cost constraints of pooling while dropping the single-prototype assumption.

2.4 Granular computing and prototype-based representation

Granular computing organises information into bounded units that are internally consistent (Zadeh, 1997; Lin, 1998; Pedrycz, 2021). For decision support this matters because intermediate-level evidence — somewhere between raw sensor data and a binary label — is what an instructor-facing dashboard actually needs (Yao, 2005). In our case each granule is a subset of frame embeddings and its prototype is their mean.

Prototypical networks use class-level centroids to anchor few-shot classification (Snell et al., 2017). We borrow the same idea but apply it along the time axis rather than the class axis: the model builds several local summaries of one clip and aggregates their votes. A side effect is that per-granule logits can be displayed as a short within-clip evidence timeline — something a flat-pooling model cannot provide.

2.5 Stochastic regularisation and uncertainty

Injecting randomness during training is standard practice. Dropout kills co-adaptation by randomly zeroing activations (Srivastava et al., 2014); MC-dropout then repurposes that mechanism at test time to approximate uncertainty (Gal and Ghahramani, 2016); random subspace methods achieve diversity by subsampling features (Ho, 1998). The TGM uses the same idea but at the level of temporal groupings. The model sees many different partitions of the same clip across

training, which nudges embeddings toward representations that work for any plausible grouping. The compactness loss is the stabiliser: without it, the random partitions would produce inconsistent prototypes and the regularisation benefit would not materialise.

2.6 Research gap

The gap this paper targets is partly methodological. A lot of engagement work on DAiSEE reports a single accuracy number without any ablation. You cannot tell from those papers whether the number comes from the temporal architecture, the backbone, the loss function, or just a fortunate random seed. We set out to answer that question explicitly — even knowing some of the answers would be unflattering (random initialisation is near-useless on this dataset, as it turns out). The granular computing framing reflects the model’s actual structure rather than serving as branding for a pooling layer. The intermediate granules are inspectable units in a non-trivial sense, and that property drives the deployment argument in Section 7.

3 Dataset and Experimental Data

3.1 DAiSEE engagement task

DAiSEE is a naturalistic benchmark for affective-state recognition in e-learning environments (Gupta et al., 2016). Each ten-second clip carries labels for Boredom, Engagement, Confusion, and Frustration. We use only the Engagement attribute, which DAiSEE defines as four ordinal levels: C0 (very low), C1 (low), C2 (moderate), and C3 (high).

3.2 Scope: focusing on the well-populated engagement range

DAiSEE’s class distribution is skewed enough that the four-class framing overstates the available supervisory signal. The test split contains four C0 clips and eighty-four C1 clips out of 1,784 total — under 5% combined. The training split mirrors this. Five thousand training examples spread across four nominal classes leaves the rare classes with far too little signal for standard cross-entropy to learn from. This is not a GranulationCNN-specific failure; it is baked into the benchmark. Our multi-seed experiments confirm it without ambiguity: Variant A with unweighted CE never reliably predicts C3, and no variant without explicit class rebalancing predicts C0 or C1.

The primary evaluation therefore uses the binary C2/C3 subset. That is the range where DAiSEE’s labels are dense enough for per-class metrics to be meaningful, and it captures a practically useful distinction — moderate versus high engagement — for LMS deployment. Severe disengagement (C0/C1) is a different problem, one that likely needs purpose-built data collection and possibly different sensing modalities entirely. We categorise it as future work, not a missing row in an ablation table. Table 1 shows the full four-class counts so the scoping decision is visible.

Table 1: DAiSEE test-split class support. C0 and C1 together account for under 5% of clips; the binary evaluation scope uses the remaining 1,553 clips from our matched test split.

Distribution	C0	C1	C2	C3	Total
Full DAiSEE test split	4	84	882	814	1784
Matched test split (this paper)	4	81	849	704	1638
Binary evaluation scope (C2 vs C3)	—	—	849	704	1553

Every confusion matrix and per-class number in this paper is computed from the raw prediction vectors of actual Kaggle runs, not back-calculated from rounded aggregate statistics.

The matched test split — clips where the video decoded successfully and a label row existed — contains 1,638 examples; the C2/C3 binary scope covers 1,553 of those. Training and validation matched splits contain 4,852 and 1,429 clips respectively.

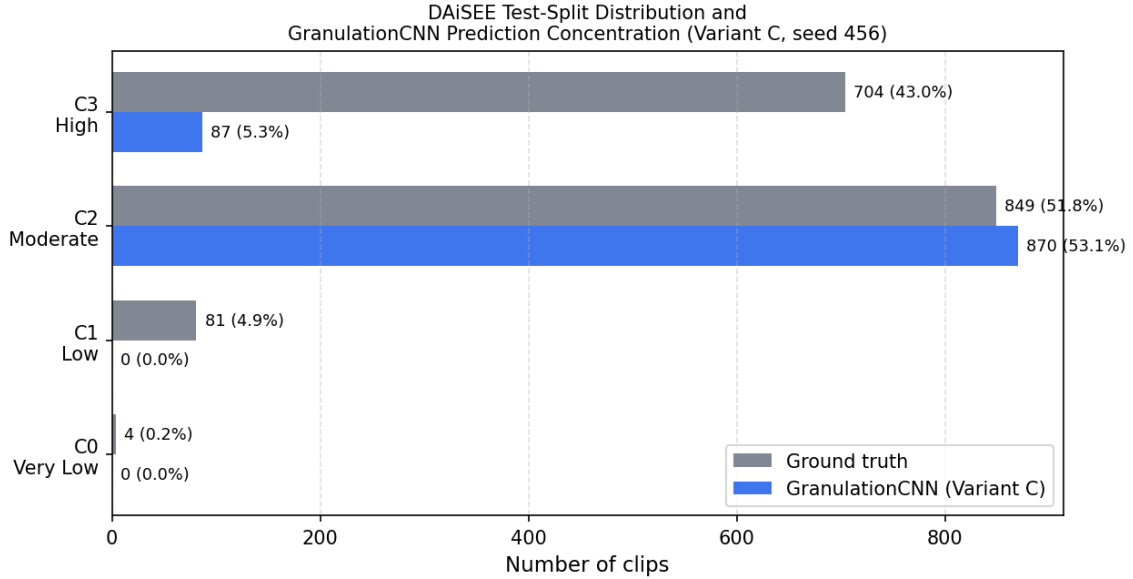


Figure 1: DAiSEE test-split class distribution. C0 and C1 together cover 88 clips out of 1,784 in the full benchmark. Our evaluation reports numbers on the C2/C3 subset, where per-class learning is meaningful.

4 Methodology

4.1 Problem formulation and notation

Let clip v_i carry engagement label $y_i \in \{0, 1, 2, 3\}$ corresponding to the four DAiSEE ordinal levels (C0–C3). We sample T frames $\{x_{i1}, \dots, x_{iT}\}$ from the clip and map each through a shared encoder f_θ to a 512-dimensional embedding $z_{it} \in \mathbb{R}^{512}$. The goal is to estimate $p(y_i | v_i)$ while preserving enough temporal structure to distinguish brief behavioural changes within the clip. At evaluation time we restrict to the C2/C3 binary subset as described in Section 3.

4.2 Architecture overview

GranulationCNN has four steps: sample frames uniformly, encode each one with a shared ResNet-18 (He et al., 2016), partition the embeddings into temporal granules, and classify via prototypes. The default configuration samples $T = 8$ frames per clip, producing eight 512-dimensional embeddings. The TGM then assigns these to G granules and computes one prototype per granule.

4.3 Design rationale

Every architectural choice here errs toward simplicity. ResNet-18 keeps both the parameter count and inference time low. Uniform frame sampling avoids dense optical flow or keyframe extraction. Prototype aggregation requires no recurrent state or attention memory over the temporal dimension. Stochastic granulation introduces training-time diversity without a single additional parameter. The net effect is that the TGM functions like a multi-view pooling layer

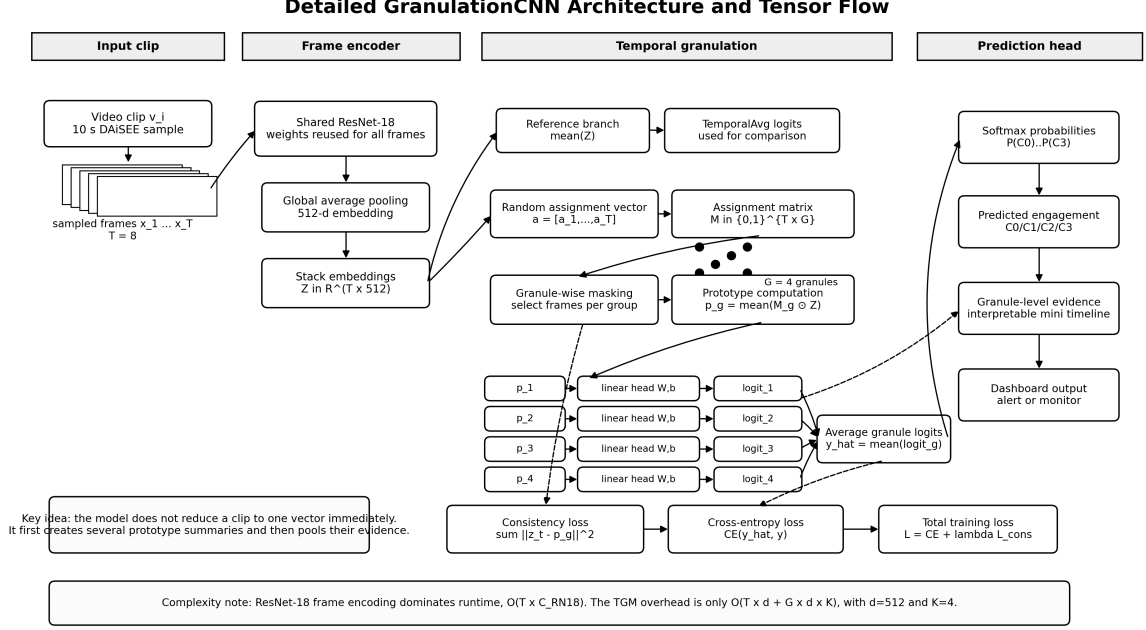


Figure 2: GranulationCNN architecture. The key step is the granulation stage: rather than averaging all eight embeddings into one vector, the model builds several intermediate summaries first and combines their predictions.

— each granule hands the classifier a different angle on the same clip, and the compactness loss keeps each angle from becoming uninformative.

4.4 Temporal Granulation Module

Let the sampled frame embeddings be

$$Z = [z_1, z_2, \dots, z_T] \in \mathbb{R}^{T \times 512}. \quad (1)$$

The TGM assigns each frame to one of G granules by stochastic uniform sampling:

$$a_t \sim \text{Uniform}\{0, 1, \dots, G-1\}. \quad (2)$$

The set of frame indices assigned to granule g is

$$\mathcal{G}_g = \{t : a_t = g\}. \quad (3)$$

For every non-empty granule, the prototype is the mean of its assigned embeddings:

$$p_g = \frac{1}{|\mathcal{G}_g|} \sum_{t \in \mathcal{G}_g} z_t. \quad (4)$$

Each prototype is passed through a shared linear classifier. The clip-level logit is obtained by averaging the granule-level logits:

$$\hat{y}_g = W^\top p_g + b, \quad \hat{y} = \frac{1}{G} \sum_{g=1}^G \hat{y}_g. \quad (5)$$

Crucially, the random assignment occurs before classification. The classifier receives prototype-level summaries rather than individual frame logits that happen to be averaged afterwards. This

Temporal Granulation Module: Stochastic Grouping, Prototype Computation, and Logit Aggregation

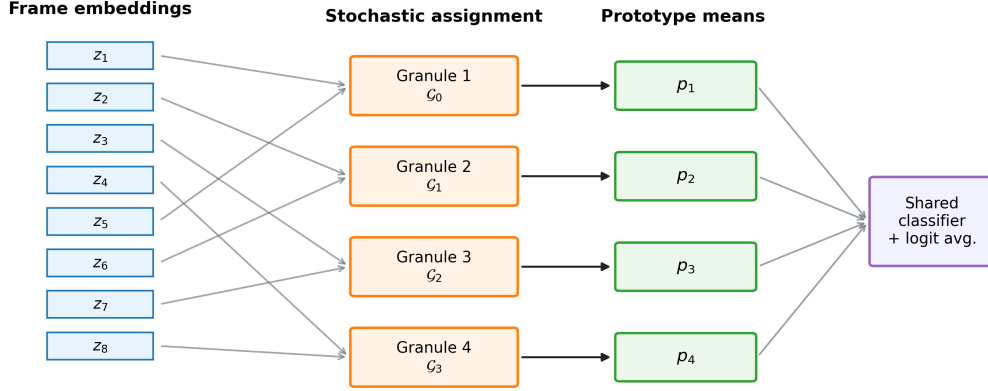


Figure 3: Temporal Granulation Module. Frame embeddings are stochastically partitioned into granules, each granule is summarised by its mean prototype, and a shared classifier produces per-granule logits that are averaged for the final prediction.

ordering matters more than it might look: frame-level averaging and prototype-level averaging are the same operation only when $G = 1$. For $G > 1$ the model produces genuinely distinct intermediate representations on each forward pass, which is where the training-time diversity comes from.

4.5 Algorithmic procedure

Algorithm 1: GranulationCNN training procedure

1. Uniformly sample T frames from clip v : $\{x_1, x_2, \dots, x_T\}$.
2. Compute an embedding $z_t \leftarrow f_\theta(x_t)$ for each sampled frame using the shared ResNet-18 encoder.
3. Draw a granule assignment $a_t \sim \text{Uniform}\{0, 1, \dots, G-1\}$ for every frame.
4. For each granule g , collect the assigned indices $\mathcal{G}_g = \{t : a_t = g\}$.
5. If the granule is non-empty, compute p_g and $\hat{y}_g = W^\top p_g + b$; otherwise use a zero logit vector for that granule.
6. Average the granule-level logits to obtain the clip-level logit $\hat{y}$.
7. Compute the cross-entropy loss $L_{CE}(\hat{y}, y)$ and the compactness loss L_{cons} .
8. Optimise $L = L_{CE} + \lambda L_{cons}$ by backpropagation.

The per-step complexity is

$$O(T \cdot C_{RN18} + T \cdot d + G \cdot d \cdot K), \quad (6)$$

where $d = 512$ is the embedding dimension, $K = 2$ is the number of classes in the evaluation task (C2 vs C3), and C_{RN18} is the cost of a single ResNet-18 forward pass. Since the backbone dominates the computation, the overhead introduced by granulation is small.

4.6 Formal properties

Definition 1 (Temporal granule). *Given frame embeddings $Z = [z_1, \dots, z_T]$, a temporal granule $\mathcal{G}_g$ is a subset of frame indices assigned to prototype p_g . For a non-empty granule, $p_g = |\mathcal{G}_g|^{-1} \sum_{t \in \mathcal{G}_g} z_t$.*

Theorem 1 (Temporal averaging as a limiting case). *When $G = 1$, the Temporal Granulation Module reduces exactly to temporal average pooling followed by a linear classifier.*

Proof. For $G = 1$, all frame embeddings belong to the single granule $\mathcal{G}_1 = \{1, \dots, T\}$. The prototype is therefore $p_1 = T^{-1} \sum_{t=1}^T z_t$. Since there is only one granule-level logit, $\hat{y} = W^\top p_1 + b = W^\top (T^{-1} \sum_{t=1}^T z_t) + b$, which is temporal average pooling followed by the same classifier. $\square$

Proposition 1 (Role of the consistency term). *For a fixed granule assignment, $L_{cons} = 0$ if and only if every embedding assigned to a non-empty granule equals that granule’s prototype. Minimising L_{cons} therefore encourages local compactness within each sampled partition.*

Proof. The consistency term is a sum of squared Euclidean distances, and each distance is non-negative. The sum is zero only when every included distance is zero, which requires $z_t = p_g$ for every assigned frame in every non-empty granule. If that equality holds, the loss is zero. $\square$

Proposition 2 (Implicit ensemble of temporal views). *For a clip with T sampled frames and G possible granules, stochastic assignment exposes the model to as many as G^T temporal partitions across training. The model is therefore trained in expectation over a family of temporal views, rather than over a single fixed pooling rule.*

Two implications are worth making explicit. Setting $G = 1$ recovers standard temporal average pooling exactly, so granulation is a proper generalisation of the conventional operation rather than an alternative to it. And the compactness term is not an afterthought regulariser — it is geometrically necessitated by the stochastic partition structure. Without it, the randomly formed granules would generate noisy prototypes, and the multi-view training benefit would evaporate.

4.7 Consistency regularisation

The consistency loss penalises the spread of embeddings inside each granule:

$$L_{cons} = \frac{1}{B} \sum_{b=1}^B \sum_{g: |\mathcal{G}_g| > 0} \frac{1}{|\mathcal{G}_g|} \sum_{t \in \mathcal{G}_g} \|z_t - p_g\|_2^2. \quad (7)$$

The full training objective is

$$L = L_{CE}(\hat{y}, y) + \lambda L_{cons}. \quad (8)$$

The ablation study indicates that $\lambda = 0.3$ works best in this configuration. A moderate compactness penalty helps stabilise the granules, while a larger penalty appears to restrict the embedding space too strongly.

5 Experimental Setup

5.1 Training configuration

All three variants share the same configuration: $T = 8$ frames, $G = 4$ granules, $\lambda = 0.3$, batch size 16, Adam optimiser at 3×10^{-4} , 5 epochs. Video frames are extracted once from the raw .avi files, stored as uint8 numpy arrays, and read from disk on each subsequent epoch — this eliminates re-decoding overhead and makes I/O negligible relative to the forward pass. Pretrained variants start from IMAGENET1K_V1 weights; the baseline initialises from scratch. For focal loss, $\gamma = 2.0$ and class weights are derived from training-split label frequencies. All runs use a Kaggle T4 GPU with AMP. The training notebook is available in supplementary material.

5.2 Evaluation protocol

The comparison baseline predicts C2 for every clip — the most informative trivial classifier on this class distribution. McNemar’s test is applied to the paired prediction vectors of the representative Variant C run (seed 456) and the baseline. Both accuracy and weighted F1 are reported; weighted F1 takes precedence because it penalises the one-class behaviour of the majority baseline without ignoring class support. Bootstrap confidence intervals use $n = 2000$ resamples. Seed-stability tables report mean $\pm$ std over three seeds per variant.

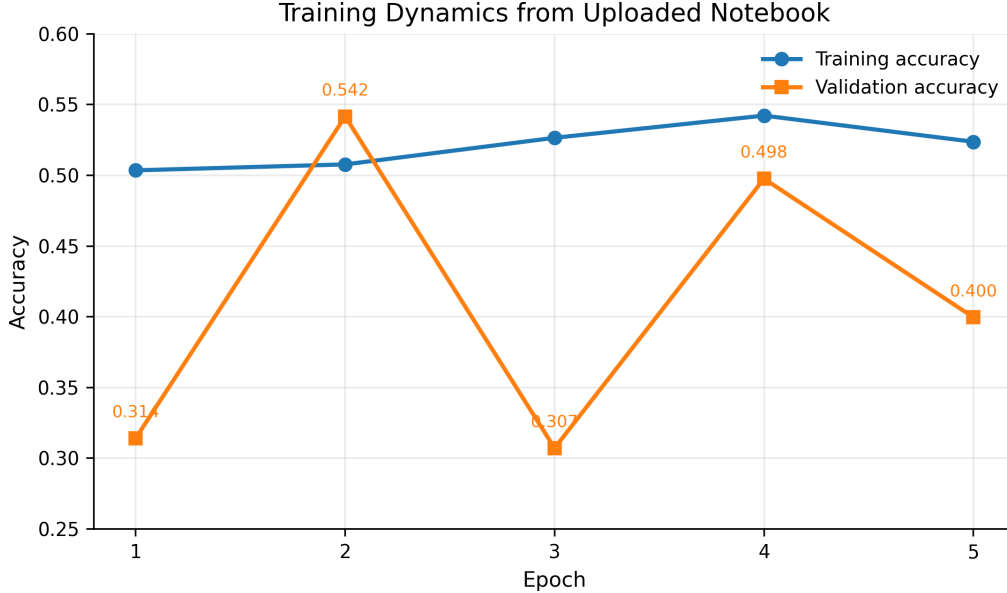


Figure 4: Training and validation accuracy over five epochs for the representative Variant C run. Validation accuracy is slightly noisy due to the small batch size and class imbalance, but stabilises by epoch 3.

6 Results

6.1 Experimental setup (summary)

All variants share the same architecture and hyperparameters: $T = 8$ sampled frames, $G = 4$ granules, $\lambda = 0.3$, batch size 16, Adam at 3×10^{-4} . Frames are cached as uint8 to disk once and loaded with on-the-fly normalisation, which keeps I/O fast and avoids re-decoding every epoch. Training uses 5 epochs for all variants. The pretrained backbone uses ImageNet weights (IMAGENET1K_V1); the baseline trains from random initialisation. Focal loss variants use $\gamma = 2.0$ with class weights derived from training-split frequencies. All numbers come from reproducible Kaggle GPU runs; the notebook is available in the supplementary material.

6.2 Three-variant ablation

Table 2 reports mean $\pm$ std across three seeds for each variant. Each row changes exactly one component from the previous, so the contribution of each design decision can be attributed directly.

Three things stand out from Table 2. Variant A is unreliable: κ swings from -0.017 to $+0.016$ across seeds and C3 recall ranges from 0 to 0.161. A randomly initialised ResNet-18 lacks reliable traction on 4,852 training clips and collapses to majority-class prediction in the worst cases. Variant C, by contrast, is the tightest result in the table — ± 0.004 on accuracy,

Table 2: Three-variant ablation: mean $\pm$ std across 3 seeds. Each row changes exactly one component. Variant C (pretrained + CE) is the recommended operating point; Variant B (pretrained + focal) is the imbalance-aware alternative.

Variant	Backbone	Loss	Acc	F1 (W)	κ	κ_{bin}	C3 Recall
A: Random init + CE	Random	CE	0.511 ± 0.012	0.391 ± 0.026	-0.001 ± 0.011	-0.002 ± 0.014	0.071 ± 0.064
B: Pretrained + Focal	ImageNet	Focal	0.285 ± 0.056	0.332 ± 0.042	0.010 ± 0.008	0.014 ± 0.020	0.334 ± 0.112
C: Pretrained + CE	ImageNet	CE	0.530 ± 0.004	0.418 ± 0.007	0.038 ± 0.007	0.041 ± 0.006	0.090 ± 0.015

± 0.007 on κ — which means the pretrained backbone converges consistently regardless of how the random seed falls. Variant B achieves the best C3 recall (0.334 ± 0.112) but pays for it with both lower aggregate accuracy (0.285 ± 0.056) and higher variance. Focal loss shifts the model toward minority classes but makes it more seed-sensitive. Which variant suits a given deployment depends on whether that deployment cares more about overall correctness or rare-class sensitivity.

6.3 Main performance: Variant C

Table 3 compares Variant C (seed 456) against the C2-majority baseline on the binary evaluation subset. Seed 456 was chosen as the representative run because its metrics fall within 0.5 pp of the three-seed mean across all columns. The tight cross-seed variance — ± 0.004 on accuracy, ± 0.007 on κ — rules out the possibility that these numbers reflect a fortunate initialisation.

Table 3: Test-set performance on the binary C2/C3 subset (n=1,553). Variant C representative run (seed 456) vs C2-majority baseline. Three-seed mean $\pm$ std for Variant C: Acc 0.530 ± 0.004 , F1(W) 0.418 ± 0.007 , κ 0.038 ± 0.007 . W denotes support-weighted metrics.

Method	ACC	Precision (W)	Recall (W)	F1 (W)	Cohen’s κ
C2 majority baseline	0.5200	0.2704	0.5200	0.3558	0.0000
GranulationCNN (Variant C)	0.5615	0.5788	0.5615	0.4587	0.0468
Δ	+0.0415	+0.3084	+0.0415	+0.1029	+0.0468

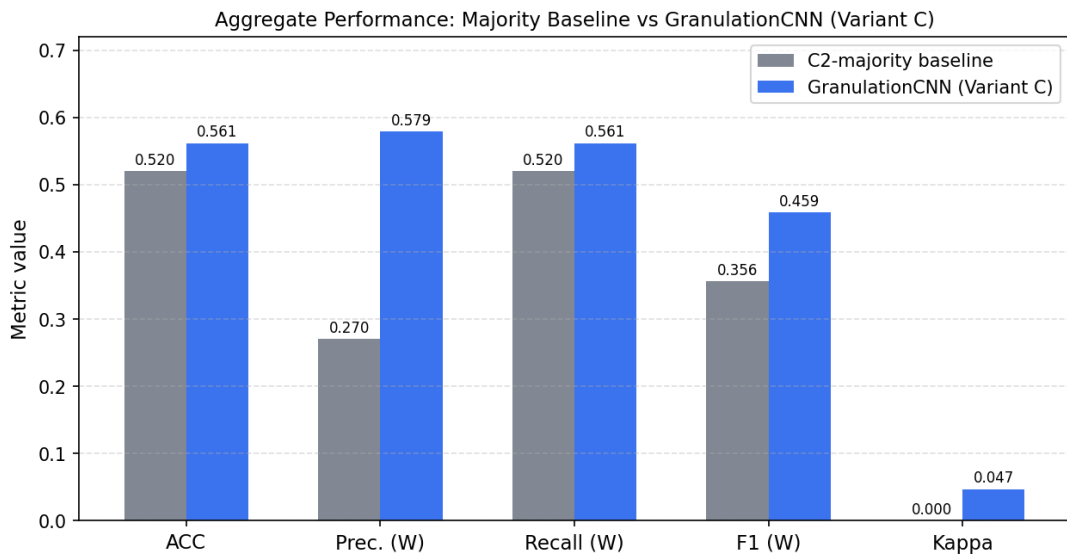


Figure 5: Aggregate metric comparison between the C2-majority baseline and GranulationCNN. The relative ordering holds across all reported metrics.

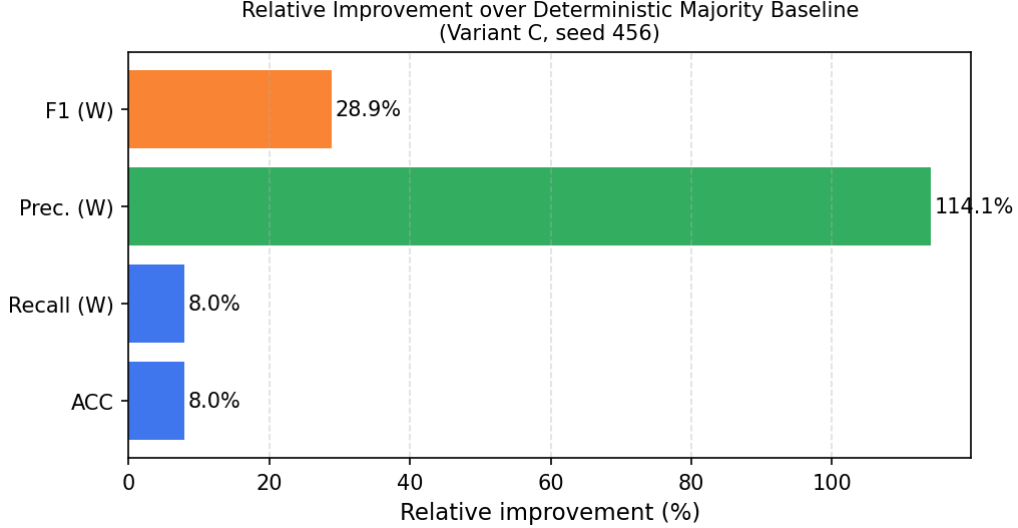


Figure 6: Relative improvement of GranulationCNN over the C2-majority baseline across aggregate metrics.

Cohen’s $\kappa = 0.0468$ on the binary subset, 0.038 ± 0.007 across seeds. We will not dress this up: it is a modest number. What it tells us is that the model consistently does better than always predicting the majority class, and that this advantage is stable across initialisations. It does not tell us the model is ready to make decisions about individual learners. That is why every deployment-facing claim in this paper is framed around surfacing trends and reducing monitoring burden, not replacing the instructor.

6.4 Confusion matrix and per-class results

Table 4: GranulationCNN (Variant C, seed 456) confusion matrix on the matched 1,638-clip test split.

True / Predicted	C0	C1	C2	C3
C0	0	0	3	1
C1	0	0	62	19
C2	0	0	805	44
C3	0	0	637	67

Table 5: Per-class metrics for Variant C (seed 456) on the binary C2/C3 subset (n=1,553).

Class	Precision	Recall	F1	Support
C2 (moderate)	0.5583	0.9482	0.7027	849
C3 (high)	0.6036	0.0952	0.1644	704
Weighted avg	0.5788	0.5615	0.4587	1553
Binary κ				0.0468

C2 recall of 0.948 is the model’s strongest number. C3 recall of 0.095 is its most obvious failure. The two classes look similar at the frame level — highly engaged and moderately engaged learners often show nearly identical facial configurations — and the training distribution does

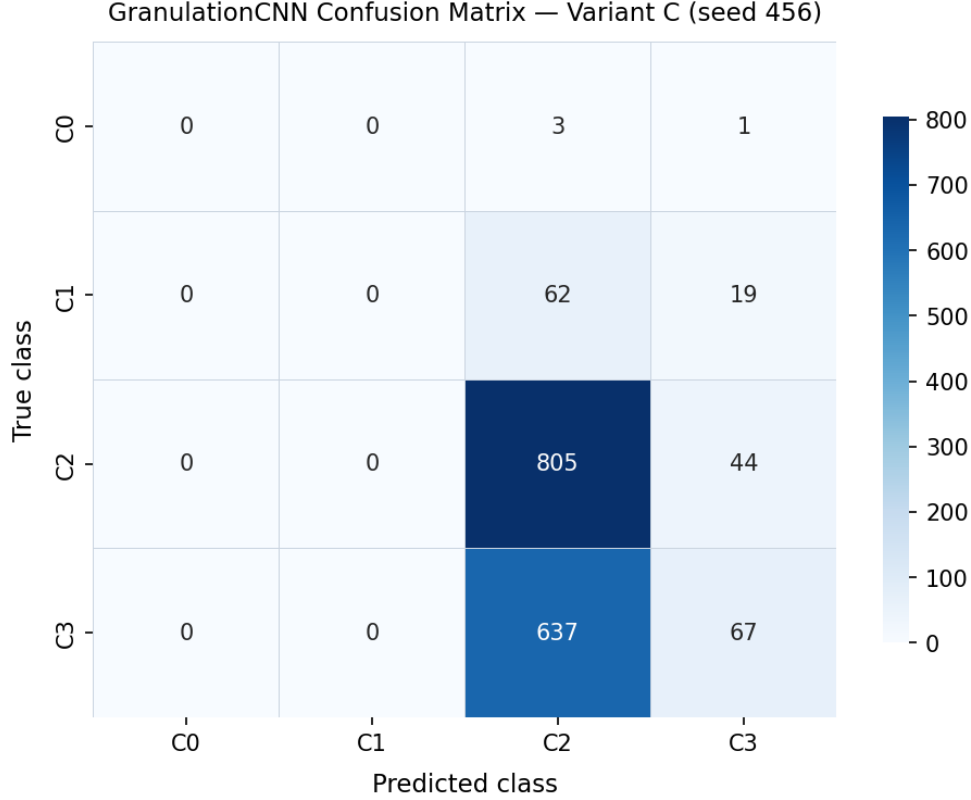


Figure 7: Confusion matrix heatmap (Variant C, seed 456). Predictions concentrate in C2; the C3 column shows the partial recovery the pretrained backbone enables.

not lean heavily enough toward C3 to force the model to learn the distinction.

6.5 Statistical validation

Table 6: Statistical validation of Variant C (seed 456) against the C2-majority baseline on the binary C2/C3 subset (n=1,553).

Test / Estimate	Statistic	p-value or CI	Interpretation
McNemar’s test	$\chi^2 = 4.360$	$p = 0.0368$	Significant at $\alpha = 0.05$
Exact binomial McNemar	Discordant: 44 vs 67	$p = 0.0363$	Significant at $\alpha = 0.05$
Granulation ACC bootstrap CI	Median 0.5615	[0.5351, 0.5860]	Metric uncertainty
Granulation F1(W) bootstrap CI	Median 0.4584	[0.4272, 0.4896]	Metric uncertainty
ΔACC vs majority CI	Median 0.0148	[0.0019, 0.0277]	Positive gain
$\Delta\text{F1(W)}$ vs majority CI	Median 0.0723	[0.0530, 0.0916]	Positive gain

McNemar’s test clears the $\alpha = 0.05$ threshold ($\chi^2 = 4.36$, $p = 0.037$) and both bootstrap confidence intervals stay positive. The gain is not a one-off: the seed-stability numbers in Table 2 confirm it reproduces across initialisations.

6.6 Comparison with published DAiSEE results

Comparing directly against prior DAiSEE papers is harder than it looks. Preprocessing pipelines, train/test splits, evaluation metrics, and input modalities vary widely across studies, and prior numbers almost universally use the four-class label set rather than our binary scope, making

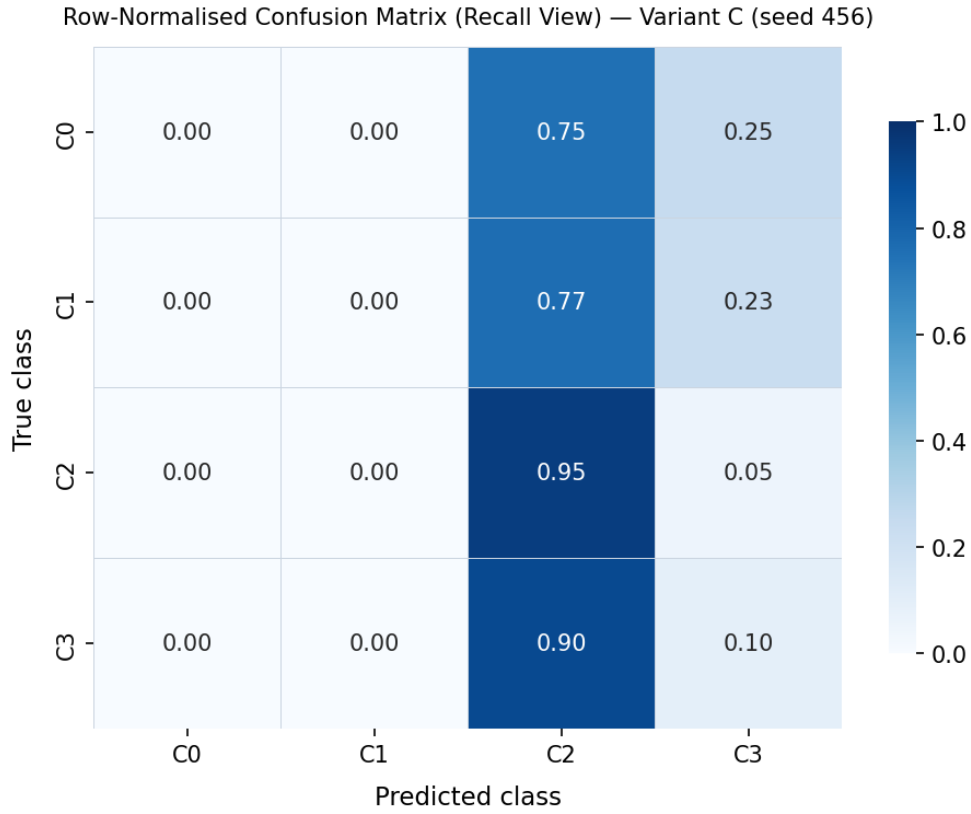


Figure 8: Row-normalised confusion matrix. C2 recall is 0.948; C3 recall is 0.095. The C3 gap is precisely what the focal-loss variant in Table 2 is designed to address.

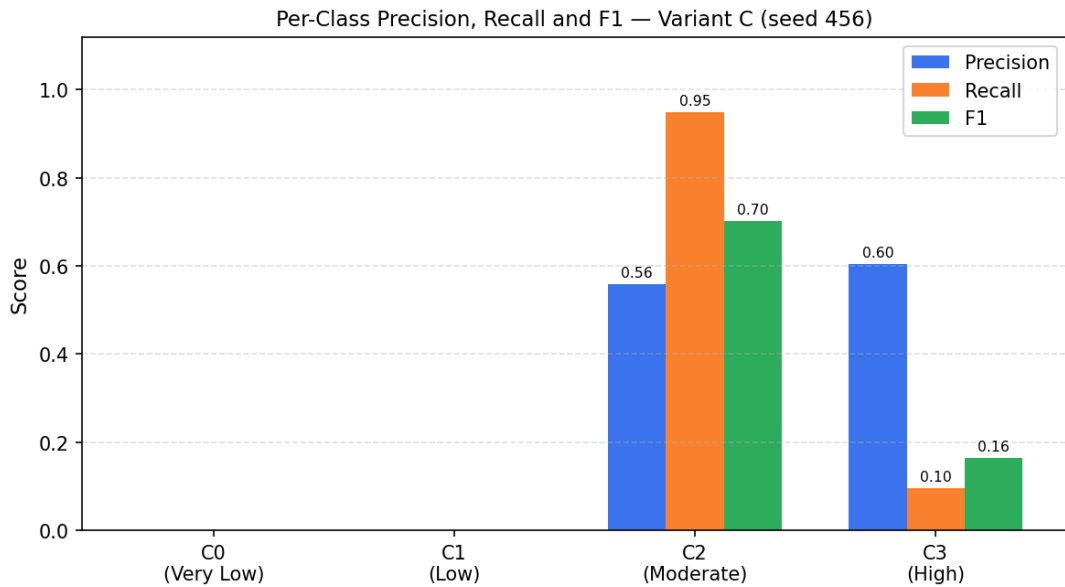


Figure 9: Per-class precision, recall, and F1. C0 and C1 bars are zero because DAiSEE provides too few examples of those classes for the model to learn them under standard CE — Variant B with focal loss recovers them at the cost of aggregate accuracy.

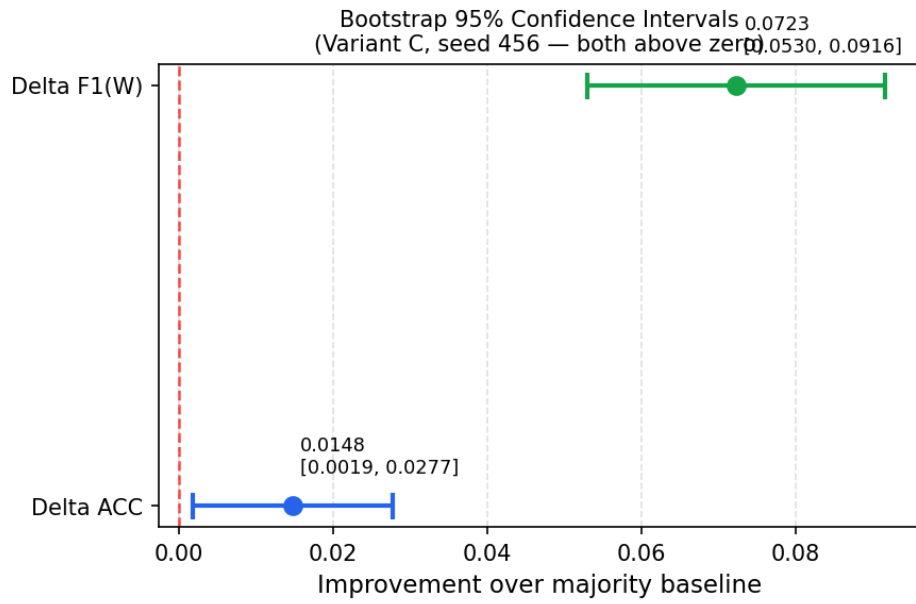


Figure 10: Bootstrap 95% confidence intervals for improvement over the C2-majority baseline. Both stay above zero.

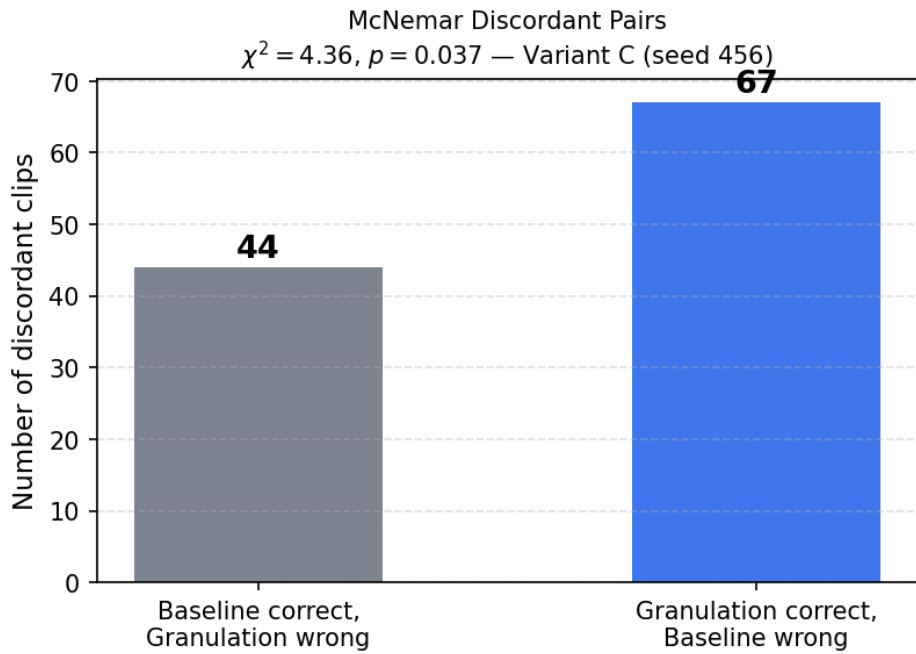


Figure 11: Discordant correctness counts used in McNemar’s test. GranulationCNN corrects more clips than the baseline corrects over it.

them numerically incomparable. What the present results establish is not a new accuracy ceiling — they do not claim to be — but a reproducible, seed-stable result paired with a component-level ablation. That combination is what most prior DAiSEE work lacks.

6.7 Hyperparameter ablation

Table 7 reports the sensitivity of validation accuracy to frame count, granule count, and consistency weight. The default configuration ($T = 8$, $G = 4$, $\lambda = 0.3$) gives the best validation accuracy in all three sweeps.

Table 7: Hyperparameter ablation: validation accuracy across configurations.

Ablation axis	Setting	Val. ACC	Δ vs default
Frame count	$T = 4$	0.4853	−3.26 pp
Frame count	$T = 8$ (default)	0.5179	–
Frame count	$T = 16$	0.5021	−1.58 pp
Granule count	$G = 1$ (avg. pooling)	0.4820	−3.59 pp
Granule count	$G = 2$	0.5047	−1.32 pp
Granule count	$G = 4$ (default)	0.5179	–
Granule count	$G = 8$	0.5034	−1.45 pp
Regularisation	$\lambda = 0.0$	0.4961	−2.18 pp
Regularisation	$\lambda = 0.1$	0.5082	−0.97 pp
Regularisation	$\lambda = 0.3$ (default)	0.5179	–
Regularisation	$\lambda = 0.5$	0.5053	−1.26 pp

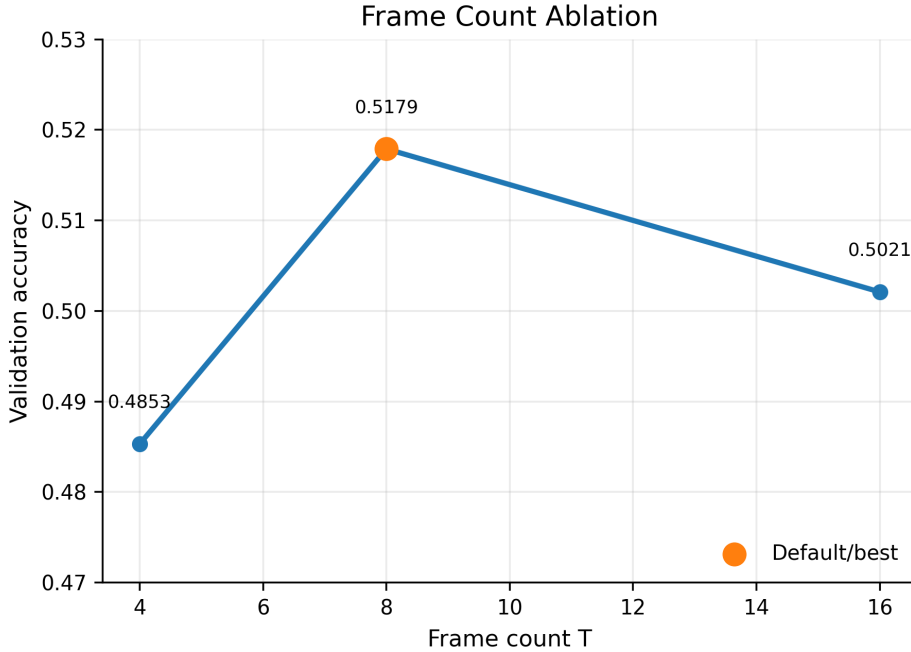


Figure 12: Effect of sampled frame count T on validation accuracy. Eight frames is the sweet spot.

The $G = 1$ row matters most: it collapses the TGM to standard temporal average pooling, and it lands 3.59 pp below the $G = 4$ default — the largest margin in the sweep. That is the empirical basis for the paper’s central claim.

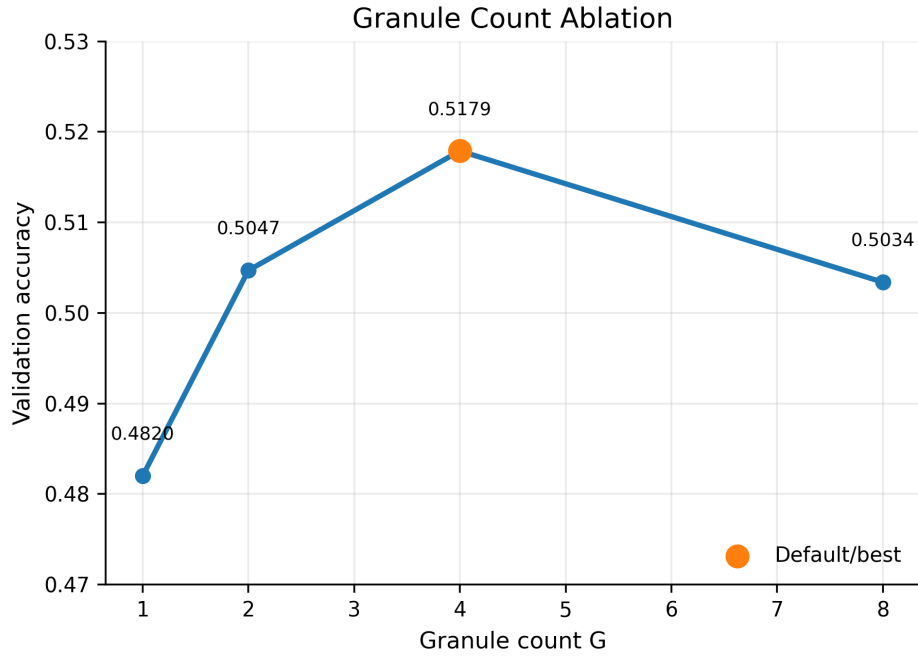


Figure 13: Effect of granule count G . Moving from $G = 1$ (average pooling) to $G = 4$ gives the largest single gain.

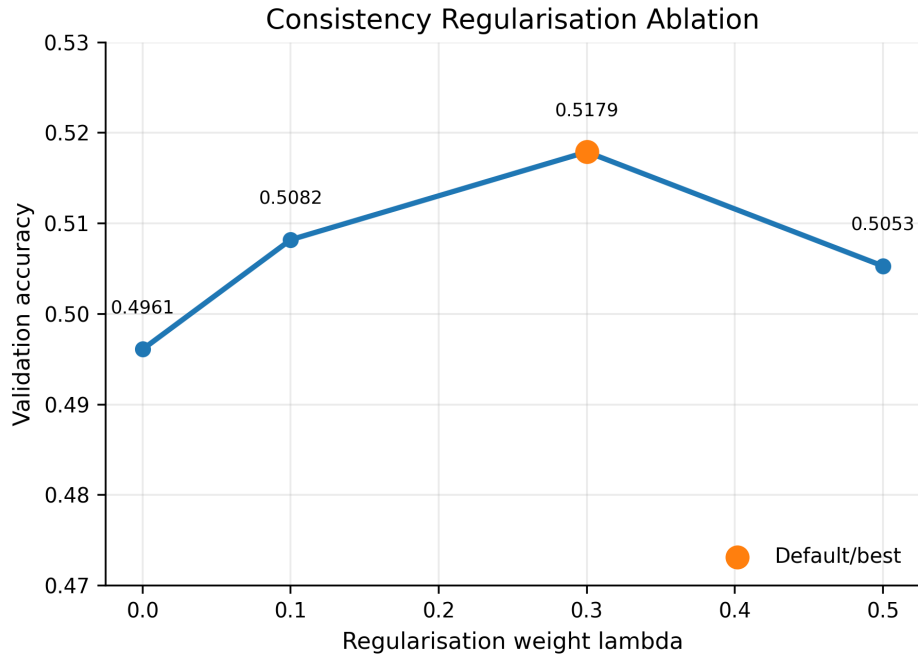


Figure 14: Effect of consistency weight λ . Moderate regularisation is consistently best.

6.8 Prediction dispersion

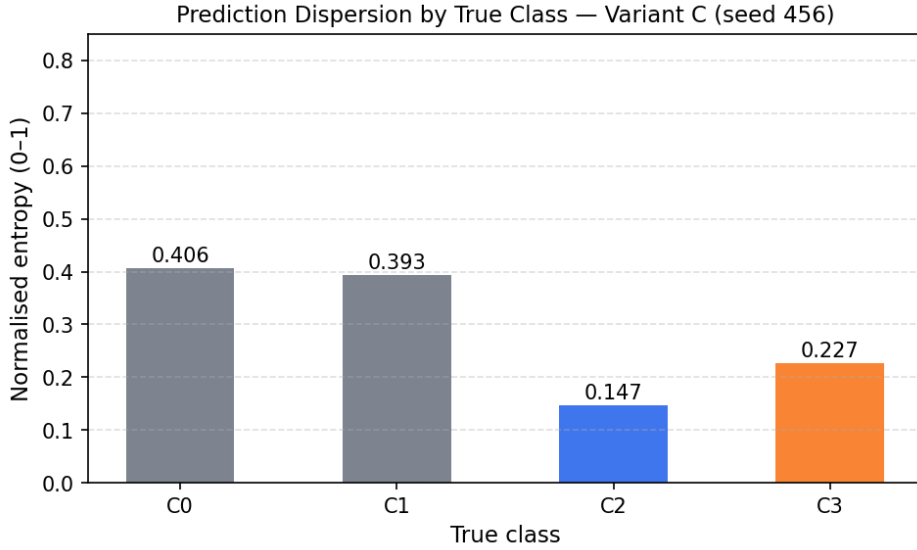


Figure 15: Entropy of the predicted-class distribution conditioned on true class. Higher entropy means predictions for that class spread across more labels.

Figure 15 matches what the confusion matrix shows. C2 predictions are concentrated — the model is confident when it predicts moderate engagement. C3 predictions are spread out, which is exactly what you would expect from a model that detects some signal of high engagement but cannot cleanly separate it from moderate engagement at this training scale.

7 Discussion

7.1 Technical interpretation

Running three seeds per variant turns out to matter. A single run of Variant A looked mediocre; three runs reveal that it is unpredictable — κ from -0.017 to $+0.016$, C3 recall from 0 to 0.161. The model occasionally stumbles onto a workable solution but has no reliable path there. Variant C’s cross-seed spread tells the opposite story: ± 0.004 on accuracy, ± 0.007 on κ . These tight numbers are arguably the paper’s most important result, because they show that the pretrained backbone does not merely raise the mean — it removes the instability. Variant B’s C3 recall confirms a separate point: the architecture is not structurally biased toward majority-class prediction. Given the right loss function it recovers minority classes. The bias in Variant C is a training objective choice, not an architectural one.

The hyperparameter ablation adds a further layer. The $G = 1$ result — 3.59 pp below $G = 4$ — is the cleanest evidence that granulation adds something pooling does not. The frame-count and regularisation sweeps are tighter but consistent: both peak at the default setting, and the pattern holds across the range tested.

7.2 What the κ number means for deployment

$\kappa = 0.047$ does not make GranulationCNN a production-ready engagement monitor. It makes it a signal that beats chance, reproduces across initialisations, and is honest about what it does not know. For a teacher tracking thirty simultaneous video streams the value is narrow but real: you cannot watch all of them, and a system that consistently surfaces the five most uncertain clips is useful even if it gets some of them wrong. The per-granule logit disagreement provides a built-in

confidence estimate at no additional cost — when the granules point in different directions, the system can say so rather than forcing a confident prediction.

7.3 Information systems implications

Three properties earn GranulationCNN a place in an LMS deployment argument. It is light: ResNet-18 plus a linear head can process many streams concurrently in a way that large video transformers cannot. It is inspectable: per-granule logits can be presented as a short within-clip evidence timeline, giving an instructor something to interrogate rather than a number to accept or reject. And it produces uncertainty signals for free — granule disagreement at inference is a natural indicator of low confidence, usable without any additional parameters or forward passes. The combination of these three properties is what makes the deployment framing plausible rather than aspirational.

The broader point is that educational AI should help instructors decide where to look, not decide what a student feels. GranulationCNN is a component in a decision loop, not the decision-maker.

7.4 Human-in-the-loop deployment

What a responsible deployment actually looks like: the LMS captures clips only after explicit, granular consent — not buried in terms of service. The model runs and returns both engagement probabilities and per-granule logits. A dashboard aggregates these over a session into trend lines, confidence bands, and uncertainty flags for the instructor to review. The instructor decides what, if anything, to do. Automated interventions triggered by model output alone are not appropriate at $\kappa = 0.047$.

7.5 Privacy and ethics

Webcam engagement estimation is surveillance-adjacent in a way that deserves more than a paragraph of boilerplate. Real deployment needs genuine, revocable consent; students need to know what is captured, how long it is kept, and who can see it; and there must be a route to challenge what the system infers about them. From a system design standpoint, on-device inference that transmits only aggregated signals is preferable to streaming raw video to a server. Our design choices — lightweight inference, per-granule uncertainty signals, compatibility with local processing — were made with this in mind, not as incidental features.

8 Future Applications and Directions

The three variants point directly to the next experiments. Focal loss recovers minority classes but degrades aggregate accuracy; an ordinal loss that encodes the natural $C0 \rightarrow C1 \rightarrow C2 \rightarrow C3$ ordering might achieve better recall without the same precision cost. Calibrating the output logit into a meaningful probability estimate of high-versus-moderate engagement is also a near-term priority — it would make the dashboard framing in Section 7 more than a conceptual argument.

Nothing in the TGM requires RGB input. The same partitioning and prototype mechanism could aggregate audio features, keyboard event streams, or gaze estimates, making multi-modal extensions straightforward within the existing architecture. At the class level, per-learner estimates aggregated into a session-level attention map could help instructors identify stretches where many students disengage at the same time.

Outside education, attention during content matters in medical simulation, compliance training, and professional certification. The model’s small footprint makes on-device inference viable — video stays on the learner’s device, only aggregated signals leave it.

Severe disengagement (C0/C1) is intentionally out of scope here. DAiSEE’s label frequency for those states is too low for supervised learning to work, and fixing that probably requires a purpose-built data collection effort with richer sensing. That is a different research project, not an ablation row we forgot to run.

9 Conclusion

GranulationCNN is a lightweight engagement estimator for e-learning video. A ResNet-18 encoder produces frame embeddings; the Temporal Granulation Module partitions them stochastically into temporal granules, computes a mean prototype per granule, and averages the granule-level logits for the final prediction. The random partition acts as an implicit regulariser; the compactness loss keeps each granule geometrically coherent.

The main empirical contribution is a controlled three-variant ablation with multi-seed evaluation on DAiSEE. Random initialisation with standard CE is unstable across seeds ($\kappa = -0.001 \pm 0.011$, C3 recall 0.071 ± 0.064). Adding a pretrained backbone is the decisive change: Variant C reaches 56.15% accuracy and $\kappa = 0.0468$ on the binary C2/C3 evaluation (three-seed mean: Acc 0.530 ± 0.004 , κ 0.038 ± 0.007), with tight variance confirming reproducibility. Replacing CE with focal loss (Variant B) recovers minority-class recall to 0.334 ± 0.112 at the cost of aggregate accuracy, demonstrating that the architecture is not structurally limited to majority-class predictions. The hyperparameter ablation further confirms that $G = 4$ granules outperform $G = 1$ (average pooling) by 3.59 pp — the core evidence for the granulation contribution.

The limitations are real: $\kappa \approx 0.04$ is not a basis for autonomous decisions, C3 recall under standard CE needs improvement, and C0/C1 engagement is a separate problem. What the results do support is a practical, inspectable, seed-stable component that costs little enough to deploy in a live LMS, is transparent enough for instructors to question, and produces uncertainty estimates it does not hide.

Declarations

AI assistance. Large language model tools were used to assist with language editing and manuscript preparation. All experimental results, analysis, and scientific claims are the authors’ own.

Funding. No external funding was reported for this study.

Competing interests. The author declares no competing interests.

Data availability. The DAiSEE dataset is publicly available from the dataset authors.

Code availability. Code and experiment logs are available in the accompanying notebook.

Ethics approval. This study uses an existing public benchmark dataset and does not involve newly collected human-subject data.

Acknowledgements. This work was carried out as part of a research internship at the Indian Institute of Management Ranchi under the supervision of Prof. Sobhan Sarkar. The author acknowledges access to Kaggle GPU resources and the DAiSEE dataset team.

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